

DEVELOPER ASSIGNMENT

Motif Design Application

Duration: 2 Hours

AI Tools: Allowed

Submission: GitHub repository link

Preferred Framework: Flutter / React Native / React / Native Android / Native iOS

Instructions

1. Build a functional prototype based on the requirements below.
2. The application should be clean, readable, and modular.
3. Mock data may be used wherever necessary.
4. The solution will be evaluated on functionality, clarity, and code quality.
5. Be prepared to explain your implementation and design decisions during review.

Question 1: Grid-Based Motif Editor

Create a grid-based motif design canvas.

Requirements

- Provide a canvas of at least 16 × 16 grid cells.
- The user must be able to:
 - tap/click individual cells to color them
 - drag to color multiple cells
 - erase selected cells
 - clear the entire canvas
- The canvas must update in real time.
- Include a visible indicator for the active tool and active color.

Output

A working grid editor where the user can create a motif by coloring cells.

Question 2: Color Selection and Editing Tools

Provide the tools required to design and edit the motif.

Requirements

- Include at least the following colors:
 - Black
 - Red
 - Blue
 - Green
 - Yellow
- The user must be able to switch the active drawing color.
- Include:
 - Draw tool
 - Eraser tool
 - Clear canvas option
- The selected tool and color should be clearly visible in the interface.

Question 3: Home Screen and Navigation

Create a home screen that acts as the main entry point of the application.

Requirements

The home screen must include the following actions:

- Create New Design
- Saved Designs
- Motif Library

Behavior

- Clicking Create New Design should open the grid-based motif editor.
- Clicking Saved Designs should show previously saved user designs.
- Clicking Motif Library should open a preset motif library.

Output

A simple home screen that allows the user to navigate through the application.

Question 4: Price Estimation

Implement a live price estimation for the motif design.

Requirements

The estimated price must update automatically based on the design.

The estimate should consider:

- number of colors used
- pattern complexity
- motif dimensions

Suggested Pricing Logic

You may use any reasonable formula, but it must be clearly implemented in the code. For example:

Final Price = Base Price + Color Charge + Complexity Charge + Size Charge

Notes

- A design using more colors should cost more.
- A denser or more intricate pattern should cost more.
- A larger motif should cost more than a smaller one.
- The price estimate must be visible while the user is designing.

Question 5: Order Handover PDF

Create a handover summary that can be shared with the production team.

Requirements

The handover document must include:

- Design name or order ID
- Final motif preview
- Colors used
- Pattern dimensions
- Complexity level or score
- Estimated price
- Any notes entered by the user

Output Format

- Generate the handover summary as a PDF
- The PDF must be downloadable from the interface

Question 6: Additional Objective - Motif Library

Provide a library of preset motifs.

Requirements

- Include at least 3 preset motif designs
- The user must be able to:
 - view the preset library
 - load a preset onto the grid canvas
 - edit the loaded preset
 - save the modified version as a new design

Question 7: Additional Objective - Saved Designs

Allow the user to save and revisit previous designs.

Requirements

- Save the current design locally
- Display previously saved designs
- Allow the user to reopen a saved design
- Allow the user to duplicate a design
- Allow the user to delete a saved design

Question 8: Additional Objective — Share Design

Provide an option to share a completed design.

Requirements

- Enable sharing or export of:
 - design image
 - design data
 - handover summary
- The sharing method may be implemented using any suitable approach supported by the chosen framework.

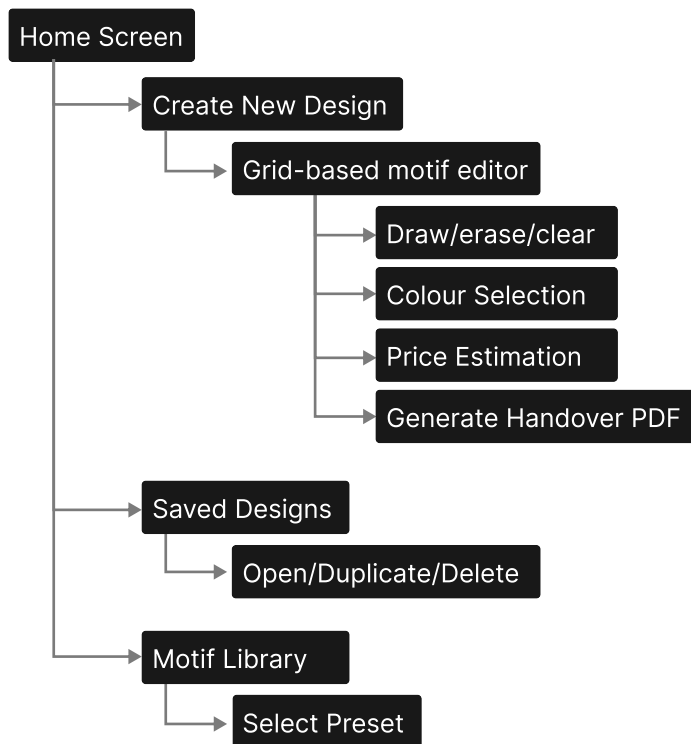
Question 8: Additional Objective — Share Design

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 - design image
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User Flow



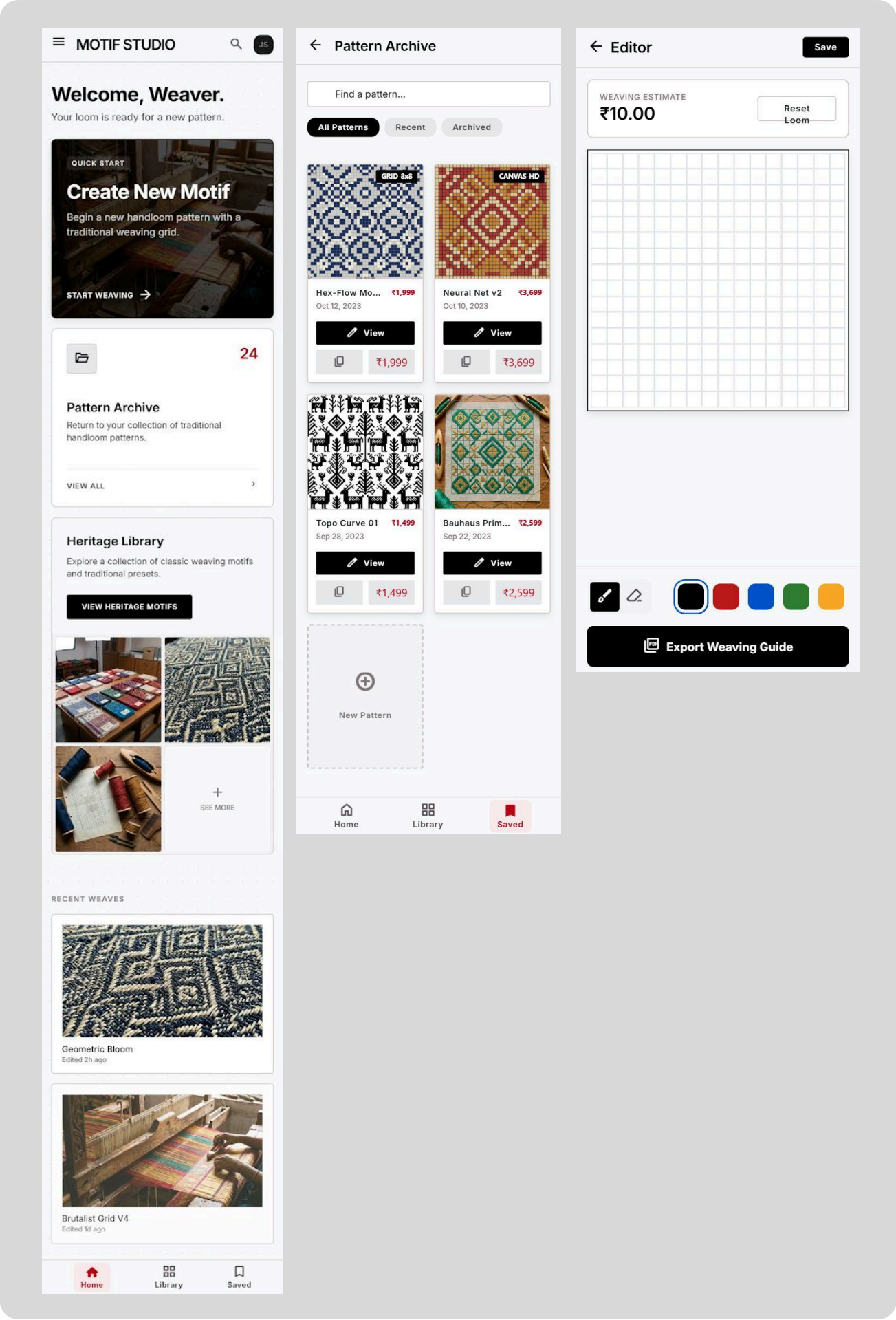
Submission

Submit the following:

- GitHub repository link
- README file with setup instructions
- Brief explanation of the price estimation logic
- Brief explanation of the data structure used for the grid canvas

Designs

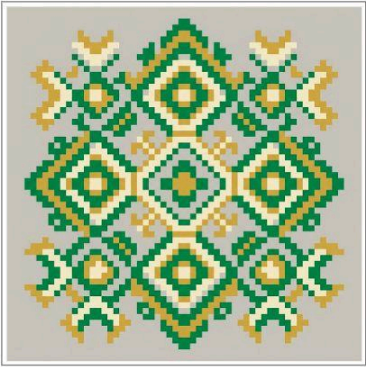
You can use these designs as reference to get started



Designs

You can use these designs as reference to get started

← Motif Summary



CURRENT CANVAS

Emerald Geometric Peak

SPECIFICATIONS

Grid Layout

16 × 16

Color Palette



COST ANALYSIS

Emerald Green (128 cells)	₹640
Gold (64 cells)	₹320
Cream (64 cells)	₹320
Pattern Complexity (High)	₹1,200
Total Estimate	₹2,480

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Library

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Profile

← MOTIF STUDIO

Heritage Library

Browse our collection of traditional handloom motifs and heritage patterns, ready for the loom.

NEW

Indigo Geometric

16X16 GRID

USE THIS PATTERN

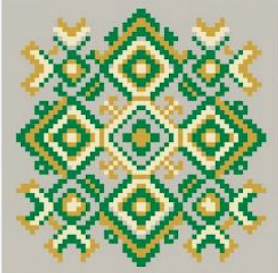
32X32 GRID

USE THIS PATTERN

Madder Rose

24X24 GRID

USE THIS PATTERN



Emerald Peak

16X16 GRID

USE THIS PATTERN

Home

Library

Saved